SEVENTH EDITION

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Calculus

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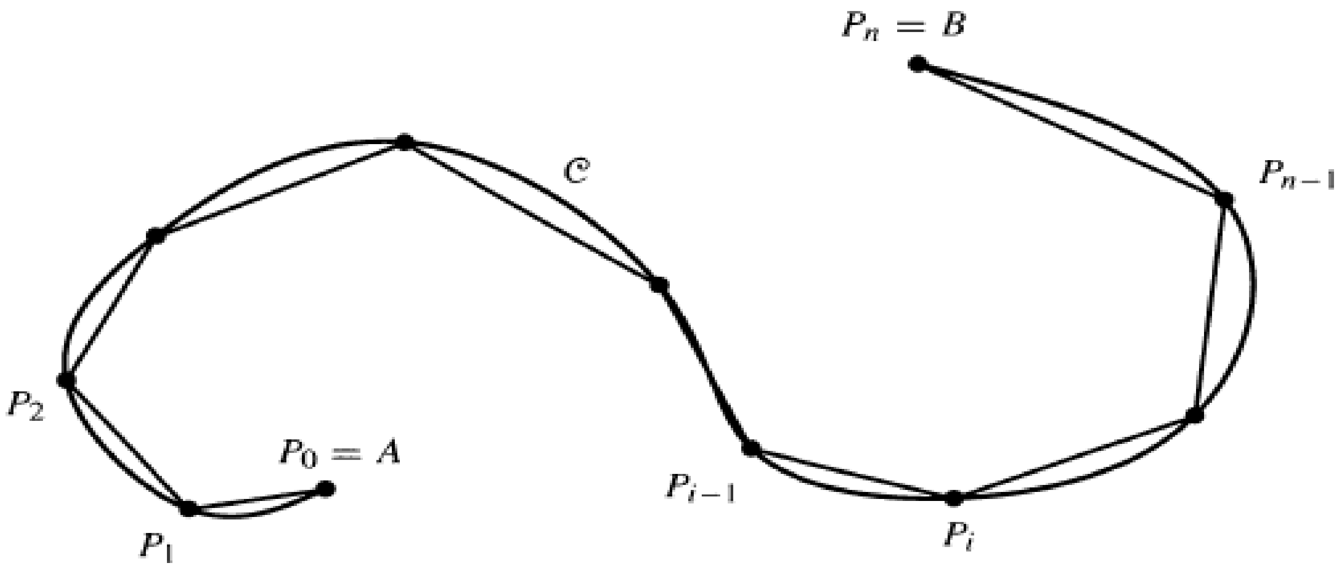
Arc Length and Surface Area

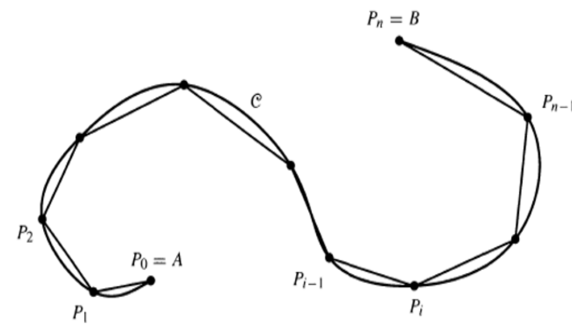
Arc Length

If A and B are two points in the plane, let $|AB|$ denote the distance between A and B, that is, the length of the straight line segment AB.



A polygonal approximation to a curve $\mathcal{C}$

The polygonal line $P_0 P_1 P_2 \ldots P_{n-1} P_n$ constructed by joining adjacent pairs of these points with straight line segments forms a *polygonal approximation* to $\mathcal{C}$, having length



$$L_n = |P_0 P_1| + |P_1 P_2| + \cdots + |P_{n-1} P_n| = \sum_{i=1} |P_{i-1} P_i|.$$

Intuition tells us that the shortest curve joining two points is a straight line segment, so the length L_n of any such polygonal approximation to $\mathcal{C}$ cannot exceed the length of $\mathcal{C}$. If we increase n by adding more vertices to the polygonal line between existing vertices, L_n cannot get smaller and may increase. If there exists a finite number K such that $L_n \le K$ for every polygonal approximation to $\mathcal{C}$, then there will be a smallest such number K (by the completeness of the real numbers), and we call this smallest K the arc length of $\mathcal{C}$.

DEFINITION

The arc length of the curve $\mathcal{C}$ from A to B is the smallest real number s such that the length L_n of every polygonal approximation to $\mathcal{C}$ satisfies $L_n \leq s$.

The Arc Length of the Graph of a Function

Let f be a function defined on a closed, finite interval $[a, b]$ and having a continuous derivative f' there. If $\mathcal{C}$ is the graph of f, that is, the graph of the equation $y = f(x)$, then any partition of $[a, b]$ provides a polygonal approximation to $\mathcal{C}$. For the partition

$$\{a = x_0 < x_1 < x_2 < \cdots < x_n = b\},$$

let P_i be the point $\big(x_i, f(x_i)\big)$, $(0 \le i \le n)$. The length of the polygonal line $P_0 P_1 P_2 \ldots P_{n-1} P_n$ is

$$\begin{aligned} L_n = \sum_{i=1}^{n} |P_{i-1} P_i| &= \sum_{i=1}^{n} \sqrt{(x_i - x_{i-1})^2 + \big(f(x_i) - f(x_{i-1})\big)^2} \\ &= \sum_{i=1}^{n} \sqrt{1 + \left(\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}}\right)^2}\, \Delta x_i, \end{aligned}$$

where $\Delta x_i = x_i - x_{i-1}$. By the Mean-Value Theorem there exists a number c_i in the interval $[x_{i-1}, x_i]$ such that

$$\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = f'(c_i),$$

so we have $L_n = \displaystyle\sum_{i=1}^{n} \sqrt{1 + \big(f'(c_i)\big)^2}\, \Delta x_i$.

Thus L_n is a Riemann sum for $\int_a^b \sqrt{1+(f'(x))^2}\,dx$. Being the limit of such Riemann sums as $n \to \infty$ in such a way that $\max(\Delta x_i) \to 0$, that integral is the length of the curve $\mathcal{C}$.

The arc length s of the curve $y = f(x)$ from $x = a$ to $x = b$ is given by

$$s = \int_a^b \sqrt{1+\left(f'(x)\right)^2}\,dx = \int_a^b \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx.$$

You can regard the integral formula above as giving the arc length s of $\mathcal{C}$ as a "sum" of **arc length elements**

$$s = \int_{x=a}^{x=b} ds, \qquad \text{where} \qquad ds = \sqrt{1 + (f'(x))^2}\, dx.$$

Figure provides a convenient way to remember this; it also suggests how we can arrive at similar formulas for arc length elements of other kinds of curves. The *differential triangle* in the figure suggests that

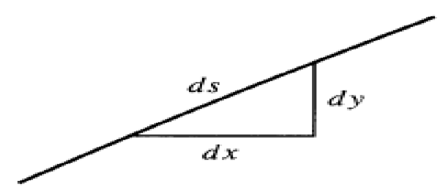
$$(ds)^2 = (dx)^2 + (dy)^2.$$

Dividing this equation by $(dx)^2$ and taking the square root, we get

$$\begin{aligned}\left(\frac{ds}{dx}\right)^2 &= 1 + \left(\frac{dy}{dx}\right)^2 \\ \frac{ds}{dx} &= \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \\ ds &= \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \sqrt{1 + (f'(x))^2}\, dx.\end{aligned}$$

A similar argument shows that for a curve specified by an equation of the form $x = g(y)$, $(c \le y \le d)$, the arc length element is

$$ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy = \sqrt{1 + (g'(y))^2}\, dy.$$



A differential triangle

Find the length of the curve $y = x^{2/3}$ from $x = 1$ to $x = 8$.

Solution Since $dy/dx = \frac{2}{3}x^{-1/3}$ is continuous between $x = 1$ and $x = 8$ and $x^{1/3} > 0$ there, the length of the curve is given by

$$s = \int_1^8 \sqrt{1 + \frac{4}{9}x^{-2/3}}\,dx = \int_1^8 \sqrt{\frac{9x^{2/3} + 4}{9x^{2/3}}}\,dx$$

$$= \int_1^8 \frac{\sqrt{9x^{2/3} + 4}}{3x^{1/3}}\,dx \qquad \text{Let } u = 9x^{2/3} + 4, \quad du = 6x^{-1/3}\,dx$$

$$= \frac{1}{18}\int_{13}^{40} u^{1/2}\,du = \frac{1}{27}u^{3/2}\Big|_{13}^{40} = \frac{40\sqrt{40} - 13\sqrt{13}}{27} \text{ units.}$$

Find the length of the curve $y = x^4 + \frac{1}{32x^2}$ from $x = 1$ to $x = 2$.

Solution Here $\frac{dy}{dx} = 4x^3 - \frac{1}{16x^3}$ and

$$\begin{aligned}1 + \left(\frac{dy}{dx}\right)^2 &= 1 + \left(4x^3 - \frac{1}{16x^3}\right)^2 \\ &= 1 + (4x^3)^2 - \frac{1}{2} + \left(\frac{1}{16x^3}\right)^2 \\ &= (4x^3)^2 + \frac{1}{2} + \left(\frac{1}{16x^3}\right)^2 = \left(4x^3 + \frac{1}{16x^3}\right)^2.\end{aligned}$$

The expression in the last set of parentheses is positive for $1 \le x \le 2$, so the length of the curve is

$$\begin{aligned}s = \int_1^2 \left(4x^3 + \frac{1}{16x^3}\right) dx &= \left(x^4 - \frac{1}{32x^2}\right)\Bigg|_1^2 \\ &= 16 - \frac{1}{128} - \left(1 - \frac{1}{32}\right) = 15 + \frac{3}{128} \text{ units.}\end{aligned}$$

(The circumference of an ellipse) Find the circumference of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

where $a \ge b > 0$. See Figure

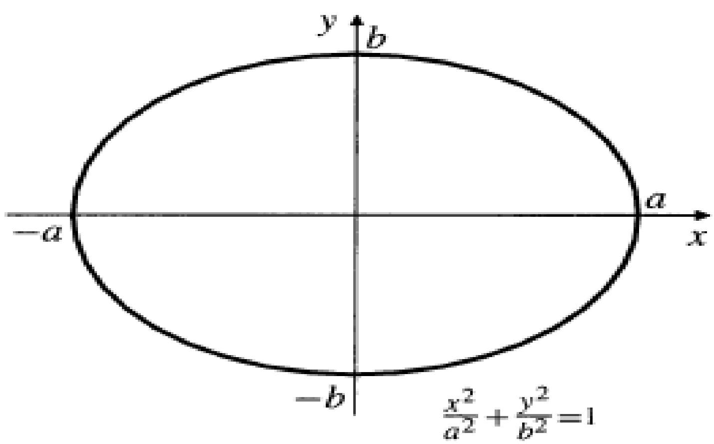
Solution The upper half of the ellipse has equation $y = b\sqrt{1 - \frac{x^2}{a^2}} = \frac{b}{a}\sqrt{a^2 - x^2}$.

Hence,

$$\frac{dy}{dx} = -\frac{b}{a}\frac{x}{\sqrt{a^2 - x^2}},$$

so

$$\begin{aligned} 1 + \left(\frac{dy}{dx}\right)^2 &= 1 + \frac{b^2}{a^2}\frac{x^2}{a^2 - x^2} \\ &= \frac{a^4 - (a^2 - b^2)x^2}{a^2(a^2 - x^2)}. \end{aligned}$$



The ellipse

The circumference of the ellipse is four times the arc length of the part lying in the first quadrant, so

$$\begin{aligned}
s &= 4\int_0^a \frac{\sqrt{a^4-(a^2-b^2)x^2}}{a\sqrt{a^2-x^2}}\,dx && \text{Let } x = a\sin t,\\
&&& dx = a\cos t\,dt\\
&= 4\int_0^{\pi/2} \frac{\sqrt{a^4-(a^2-b^2)a^2\sin^2 t}}{a(a\cos t)}\,a\cos t\,dt\\
&= 4\int_0^{\pi/2} \sqrt{a^2-(a^2-b^2)\sin^2 t}\,dt\\
&= 4a\int_0^{\pi/2} \sqrt{1-\frac{a^2-b^2}{a^2}\sin^2 t}\,dt\\
&= 4a\int_0^{\pi/2} \sqrt{1-\varepsilon^2\sin^2 t}\,dt \text{ units,}
\end{aligned}$$

where $\varepsilon = (\sqrt{a^2-b^2})/a$ is the *eccentricity* of the ellipse.
Note that $0 \le \varepsilon < 1$. The function $E(\varepsilon)$, defined by

$$E(\varepsilon) = \int_0^{\pi/2} \sqrt{1-\varepsilon^2\ \sin^2 t}\,dt,$$

is called the **complete elliptic integral of the second kind**. The integral cannot be evaluated by elementary techniques for general ε, although numerical methods can be applied to find approximate values for any given value of ε. Tables of values of $E(\varepsilon)$ for various values of ε can be found in collections of mathematical tables. As shown above, the circumference of the ellipse is given by $4aE(\varepsilon)$. Note that for $a = b$ we have $\varepsilon = 0$, and the formula returns the circumference of a circle; $s = 4a(\pi/2) = 2\pi a$ units.